

Jackson Parrack

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EDUCATION

Georgia Institute of Technology Atlanta, GA <i>Master of Science in Computer Science, Computing Systems</i>	Expected May 2029
Saint Louis University St. Louis, MO <i>Bachelor of Science in Computer Engineering, Minor in Computer Science</i>	Aug. 2022 – May 2026 GPA: 3.93/4.0
Sogang University Seoul, South Korea <i>Exchange Program</i>	Feb. 2024 – Jun. 2024

EXPERIENCE

Mastercard <i>Software Engineer I</i>	O'Fallon, MO Aug. 2026 – Present
- Architecting and leading a Python MCP service on Kubernetes giving 140+ employees agentic access to Oracle databases, piloted program-wide as the enterprise MCP blueprint: read-only SQL guardrails with query budgets and timeouts, OAuth that inherits each user's database permissions, vault-injected secrets, Jenkins CI/CD, and Splunk observability. - Taking over development of a Go-based simulator that tests the full payment lifecycle end-to-end, growing it into the program's in-house integration testing platform to replace a costly vendor tool. - Developing a modernized clearing platform with Java Spring microservices on Kubernetes, replacing mainframe technology with services that scale horizontally to 10M transactions/day and integrate with a new settlement platform. - Researching high-performance networking (Homa, eBPF/XDP, RDMA) with distinguished engineers, building VM testbeds and working in the Linux kernel networking stack to benchmark each against TCP for data center deployment.	
DataServ <i>DevOps Intern</i>	St. Louis, MO May 2025 – Aug. 2025
- Wrote Python automation scripts to migrate 3,000+ SQL stored procedures across 71 client environments into Bitbucket, enabling code reviews and rollbacks for previously unversioned code. - Converted time-based jobs to event-driven triggers for 7 clients, contributing to a company-wide effort that cut client data turnaround from 24 hours to under 6.	

PROJECTS

Rapid Photonic Innovation Device (RAPID) <i>ECE Capstone Design Award Winner</i> GitHub <i>Converted a Blu-ray drive into a ~\$500 FPGA-controlled laser lithography system with ~0.1mm resolution</i>	Aug. 2025 – Apr. 2026
- Owned all software and supported FPGA integration on a 4-person team, spanning embedded C, VHDL, and Python. - Built bare-metal C firmware on a Zynq SoC for motor and laser control, syncing laser timing to disc rotation. - Designed a custom UART protocol with error checking and flow control to stream patterns from the PC. - Built a real-time Python GUI for live pattern visualization, manual control, and system telemetry.	

LEADERSHIP

IEEE Student Branch <i>Vice President</i>	Aug. 2025 – May 2026
- Revived a dormant student chapter, growing membership from under 10 to 30+ through outreach and events. - Captained the team that won 1st place at the 2026 IEEE Region 5 Cyber Challenge.	
SLU Esports <i>President and Teams Coordinator</i>	Aug. 2024 – May 2026
- Grew membership to 70+ and coordinated competitive rosters across 10+ titles. - Directed a 6-person executive board, delegating ownership to each officer and coordinating their work across the club.	

SKILLS

Programming Languages: C, C++, Python, Java, Go, SQL, VHDL
Tools/Platforms: Spring, Kubernetes, Docker, Jenkins, Splunk, MCP, Git, Linux
Hardware: FPGA, ARM & AVR Microcontrollers
Spoken Languages: Spanish (Advanced), Korean (Intermediate)